

Exp. No: 3.

DETERMINATION OF VOLTAGE IN CIRCUIT

USING NODAL ANALYSIS.

10/2/2026

AIM:

To determine the Voltage in the Circuit using nodal analysis both theoretically and Practically for a given Dc Circuit.

APPARATUS REQUIRED

SL.No	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated power Supply (RPS)	(0-30V)	1
2.	Multimeter	-	1
3.	Resistors	470 Ω , 330 Ω	3, 2
4.	Bread board	-	1

PROCEDURE:

- i) Given Connections as per the Circuit diagram
- ii) Switch ON the Supply, Vary the RPS and Set a Particular Input Voltage
- iii) Note down the Readings of ammeter and Voltage and tabulate them.

vi) using the tabulated Values, Verify Kirchhoff's laws Practically & Verify it theoretically.

MARKS SPLIT-UP:

S.No	Description	Marks allotted	obtained.
1.	Experimental Setup/ Connections	20	20
2.	Observation / readings	20	20
3.	Calculations / results	20	20
4.	Viva	20	20
5.	Records	20	20
Total			100

RESULT

Thus, the Nodal analysis is Verified Practically & theoretically. The resultant Voltages for 2 V Supplies:

8/6
4/12/28

a) The Voltage V_1 is 0.63 V

b) The Voltage V_2 is 0.45 V

c) The Voltage V_3 is 0.18 V

TABULAR COLUMN:

Parameters	Theoretical	Practical
$V_A = V_1$	0.63V	0.66V
$V_B = V_2$	0.45V	0.45V
$V_A - V_B = V_3$	0.18V	0.19V

CALCULATIONS:

At node A:

By KCL:

$$\frac{V_A - 2}{470} + \frac{V_A}{330} + \frac{V_A - V_B}{470} = 0$$

$$\frac{33V_A - 66 + 47V_A + 33V_A - 33V_B}{(15)(570)} = 0$$

$$113V_A - 33V_B = 66 \quad \text{--- (1)}$$

At node B:

$$\frac{V_B - V_A}{470} + \frac{V_B}{330} + \frac{V_B}{470} = 0$$

$$33V_B - 3V_A + 47V_B + 33V_B = 0$$

$$113V_B - 3V_A = 0$$

$$V_B = \frac{33}{113} (V_A) \quad \text{--- (2)}$$

Sub (2) in (1)

$$113V_A - 3 \left(\frac{33}{113} V_A \right) = 66$$

$$113V_A - \frac{99}{113} V_A = 66$$

$$V_B = 0.126V$$

$$V_A = \frac{113}{33} V_B$$

$$= \frac{113}{33} \times 0.12$$

$$= 0.63V$$

$$V_1 = V_A = 0.63V$$

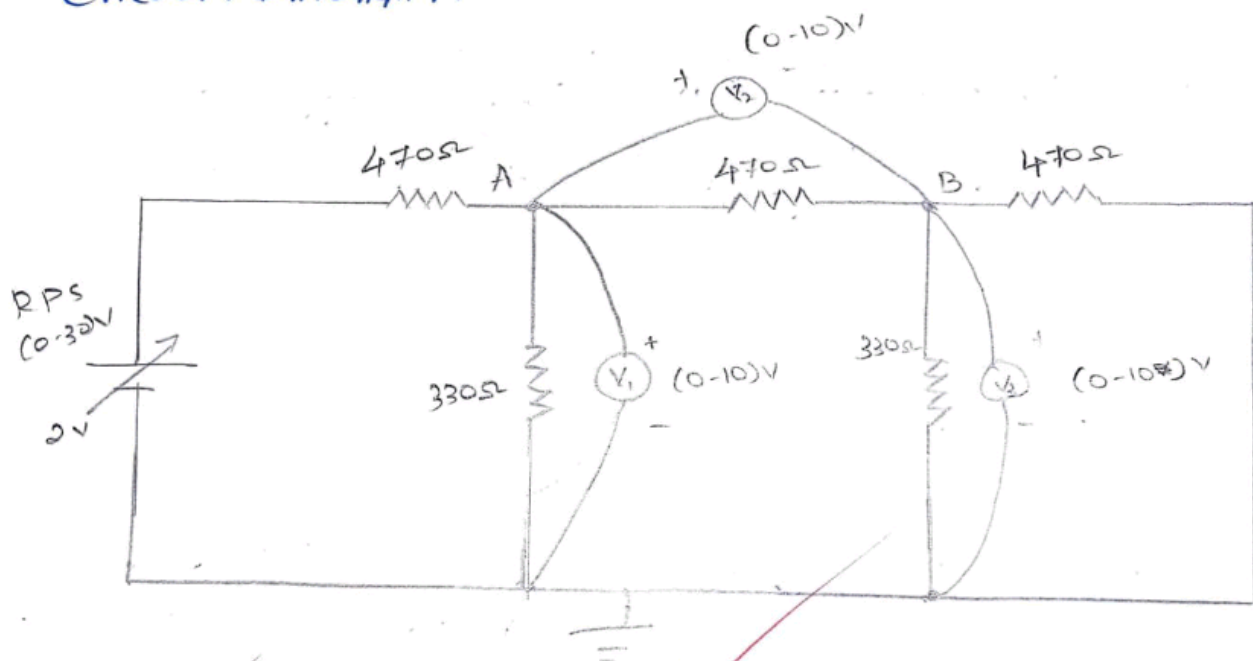
$$V_2 = V_B = 0.12V$$

$$V_3 = V_A - V_B =$$

$$0.63 - 0.12$$

$$= 0.45V$$

CIRCUIT DIAGRAM:



Nodal Circuit